

PAPER_821: RIAF CRP, IceCube Neutrino Background, and LLAGN Fokker-Planck UQFF

Unified Quantum Field Framework — Whitepaper 821

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Abstract

This paper derives the Radiatively Inefficient Accretion Flow (RIAF) Cosmic Ray Proton (CRP) acceleration and high-energy neutrino emission terms for the Quadriadic UQFF. A maximally spinning BH ($a_* = 0.9375$, $M_{BH} = 10^8 M_\odot$, $\dot{M} = 10^{-2} L_{Edd}/c^2$) produces $L_\nu \approx 6 \times 10^{38}$ erg/s in the 1-100 TeV band, consistent with the IceCube diffuse neutrino background from the LLAGN population. The Fokker-Planck CRP acceleration mechanism with $\tau_{acc} = 1/(2K)$, where K is the spatial diffusion coefficient, governs the CRP energy spectrum and neutrino SED. RIAF outflow CRPs dominate over inflow CRPs in the effective acceleration zone.

1. Introduction

LLAGN (Low-Luminosity AGN) with accretion rates $f_{Edd} \ll 0.01$ spend most of their accretion energy in a geometrically thick, hot RIAF (alternatively termed ADAF — Advection-Dominated Accretion Flow). Unlike standard thin disks, the RIAF is optically thin, ion-pressure-dominated, and has sub-relativistic inflow velocities. These conditions create a natural particle accelerator for cosmic ray protons.

2. RIAF Parameter Configuration

- **BH spin:** $a_* = 0.9375$ (near-maximal)
 - **BH mass:** $M_{BH} = 10^8 M_\odot$
 - **Accretion rate:** $\dot{M} = 10^{-2} \cdot L_{Edd}/c^2$
 - **Gravitational radius:** $r_g = GM_{BH}/c^2 \approx 1.5 \times 10^{11}$ km
 - **CRP residence:** $> 10^3 r_g/c$ (long enough for efficient acceleration)
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3. CRP Fokker-Planck Acceleration

The cosmic ray proton distribution evolves under the Fokker-Planck equation:

$$\frac{\partial N}{\partial t} = \frac{\partial}{\partial p} \left[D_{pp} \frac{\partial N}{\partial p} \right] - \frac{\partial}{\partial p} [\dot{p}_{loss} N] - \frac{N}{\tau_{esc}}$$

where $D_{pp} \propto p^2/(2\tau_{acc})$ is the momentum diffusion coefficient.

The acceleration timescale:

$$\tau_{acc} = \frac{1}{2K}$$

where $K = D_{pp}/p^2$ is related to the spatial diffusion and the Alfvén wave turbulence spectrum.

4. CRP Injection Rate

The CRP injection rate is proportional to the magnetic field variation:

$$\dot{N}_{inj} = \eta_{inj} \cdot \max(1, \beta_{mag}) \cdot \frac{|DB'/Dt|}{|B'|}$$

where $\beta_{mag} = U_{th}/U_{mag}$ is the thermal-to-magnetic energy ratio, and B' is the fluctuating magnetic field. For RIAF with $\beta_{mag} \approx 5-20$:

$$\dot{N}_{inj} \propto v_A \cdot r^{-2}$$

UQFF Layer 2:

$$g_{L2,acc} = 1/\tau_{acc} \approx 10^5 \text{ m/s}^2$$

5. CRP Outflow Dominance

RIAF outflow CRPs vs inflow CRPs: - **Inflow**: CRPs accreted into BH within $< 10r_g$ — lost - **Outflow**: CRPs carried by MHD-driven winds at $> 10^3 r_g/c$ residence

The effective outflow power:

$$P_{CRP,out} \approx 3.41 \times 10^{44} \text{ erg/s}$$

This exceeds the AGN bolometric luminosity for the given accretion rate, confirming that RIAF is an efficient mechanical energy reservoir.

UQFF Layer 3:

$$g_{L3,CRP} = \eta_{inj} \cdot B^2/r \approx 10^{27} \text{ m/s}^2$$

6. Neutrino SED — 1-100 TeV

The neutrino luminosity from pp , $p\gamma$ interactions in the RIAF corona:

$$L_\nu(\epsilon_\nu) = \int \sigma_{pp}(E_p) \cdot n_{target} \cdot N_{CRP}(E_p) \cdot \frac{c}{3} \cdot K_\nu dE_p$$

For $a_* = 0.9375$, $\dot{M} = 10^{-2}$:

$$L_\nu(1\text{-}100 \text{ TeV}) \approx 6 \times 10^{38} \text{ erg/s}$$

UQFF Layer 4:

$$g_{L4,\nu} = L_\nu/r^3 \approx 6 \times 10^{-7} \text{ m/s}^2$$

7. IceCube Diffuse Background Match

The total diffuse IceCube neutrino intensity from the LLAGN population:

$$E_\nu^2 \Phi_\nu = \int_0^{z_{max}} \frac{c}{4\pi H_0} \cdot \frac{dn_*}{dL_{H\alpha}} \cdot \epsilon_\nu L_{\epsilon_\nu} dz$$

Two-population model: 1. NGC 1068-like sources (bright, high f_{Edd} , $\sim 0.1 \text{ erg/s/cm}^2$ at IceCube threshold) 2. LLAGN RIAF population (this paper, high-spin low-luminosity, diffuse background)

The LLAGN RIAF component accounts for $\sim 20\%$ – 40% of the IceCube diffuse isotropic background at 1–100 TeV.

8. Spin Dependence of Neutrino Emission

Higher spin a_* directly increases L_ν :

$$L_\nu \propto a_*^{2/3} \cdot \dot{M}^{3/2}$$

For $a_* = 0.9375$ vs $a_* = 0.5$:

$$L_\nu(0.9375)/L_\nu(0.5) \approx 1.7$$

9. Summary

RIAF systems with near-maximal spin ($a_* = 0.9375$) generate $L_\nu \approx 6 \times 10^{38} \text{ erg/s}$ in the 1–100 TeV IceCube band via CRP Fokker-Planck acceleration. Outflow CRPs (residence $> 10^3 r_g/c$) dominate neutrino production. The LLAGN RIAF population contributes 20%–40% to the IceCube diffuse background. Acceleration timescale $\tau_{acc} = 1/(2K)$, injection rate $\propto |DB'/Dt|/|B'|$, and neutrino luminosity $\propto a_*^{2/3}$ are now formal Quadriadic UQFF Layers 2–4 parameters.
